

Delay–Energy Aware Dynamic Master Head Selection in Intelligent Healthcare IoT Networks

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Abstract—Over the years, the rapid growth of healthcare Internet of Things (IoT) systems demands intelligent networking solutions capable of supporting latency-sensitive signal processing while maintaining energy efficiency. Conventional cloud-centric architectures often suffer from additional delay and resource overhead, making them unsuitable for time-critical real-time biomedical applications. However, fog computing architecture provides solution to cloud-based processing but still possess challenges such as dynamic environment, and heterogeneity. To address these challenges, this paper proposes a dynamic master head selection framework in a heterogeneous fog-assisted healthcare IoT environment. The selection problem jointly minimizes end-to-end delay and fog energy consumption using Pareto optimization strategy. Extensive simulation results demonstrate that the proposed approach achieves up to 35% reduction in additional total delay with 60 – 70% improvement in energy efficiency compared to static fog-IoT environment and cloud-centric architecture. The proposed framework is suitable for healthcare centers with near-edge processing devices, supporting intelligent distributed computing for EEG signal processing.

Index Terms—Internet of Things, Fog computing, Intelligent Head Selection, Pareto optimization, Biomedical Signal Processing, Delay, Energy.

I. INTRODUCTION

The rapid evolution of wireless communication technologies and the Internet of Things (IoT) has significantly transformed modern healthcare systems by

enabling continuous patient monitoring, real-time physiological data acquisition, and intelligent clinical decision support. Among various biomedical signals, Electroencephalogram (EEG) data play a crucial role in diagnosing and managing neurological disorders, brain–computer interface applications, and cognitive health assessment [1]. However, EEG devices in IoT networks generate large volumes of time-critical healthcare data, posing substantial challenges in terms of transmission delay, energy consumption, and scalable processing. Traditional cloud-centric healthcare architectures provides large computational resources for processing these data and remote analysis but suffer from additional end-to-end latency, high bandwidth, and high energy overhead due to long-distance data transmission and centralized processing [2]. To address these limitations, fog computing paradigms have emerged as promising solutions. In fog-assisted healthcare IoT systems, intermediate fog nodes provide localized processing, temporary storage, and rapid response, while the cloud performs long-term analytics and permanent storage [3]. In recent times, many researchers have adopted fog computing paradigm for intelligent computing in EEG healthcare systems, considering heterogeneous fog devices [4] [5], still efficiently coordinating in dynamic environments remains a critical research problem. Therefore, this work focuses on a EEG-fog-cloud network for intelligent signal processing in a dynamic fog environment.

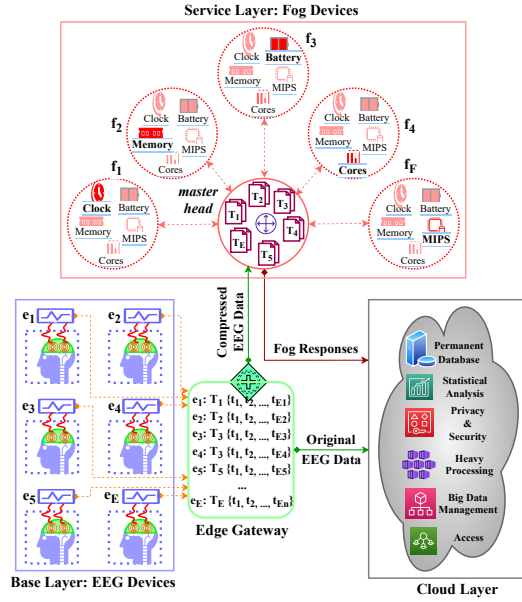


Fig. 1. Shows abstract view of 3-layer fog-IoT architecture for proposed work.

A. Motivation and contributions

The motivation for this work stems from three key observations. First, many EEG-based healthcare services such as seizure detection, sleep disorder, and cognitive monitoring etc. demand near real-time response [6] [7] [8]. Even moderate delays can compromise clinical effectiveness. Second, Fog devices are often resource-limited and energy-constrained. Prolonged operation under suboptimal task allocation can significantly reduce network span. Third, Unlike static infrastructures, fog networks are inherently dynamic, with varying node availability and placement. Therefore, static fog environments are less effective in parallel and distributed processing [9]. These challenges highlight the need for a dynamic and intelligent master head selection mechanism that jointly considers communication delay and energy consumption. To address the above challenges, this paper presents a dynamic master head selection framework with the following key contributions.

- A three-layer architecture is designed, comprising multiple EEG devices, heterogeneous fog devices, an edge gateway, and a cloud infrastructure.
- We propose the master head selection problem that jointly minimizing end-to-end delay and energy consumption in dynamic fog environment. We adopt Pareto based optimization strategy to select a Master Head (MH) among non-dominated fog

candidates.

- A comprehensive simulation framework is constructed using a benchmark dataset to evaluate performance of this work over 100 iterations. Moreover, we demonstrate the result analysis by varying number of fog devices and task sets, and compare against baseline cloud architecture and static fog-IoT environment..

II. SYSTEM MODEL

The proposed healthcare IoT system architecture comprises of three layers - base layer, service layer, and cloud layer, interconnected through an edge gateway and a master head, as illustrated in Fig. 1. At the base layer, multiple EEG devices continuously acquire raw brain signals from patients and generate task sets associated with physiological monitoring. Each EEG device encapsulates its sensed data into computational tasks and forwards them to the edge gateway. The edge gateway acts as an interface between EEG devices and the fog infrastructure. It aggregates incoming EEG task streams and forwards toward the service layer. The gateway also performs EEG data compression locally, thereby reducing redundant transmissions and improving network efficiency [10]. The service layer consists of multiple heterogeneous fog devices characterized by distinct computational capabilities such as clock frequency, memory, number of cores and million instructions per second (MIPS), and residual energy levels. These fog devices are responsible for collaborate and distributed processing and decision-making for EEG healthcare data. Among them, a dynamically designated fog device, denoted as MH, is responsible for three main services - collecting tasks from the gateway, distributing workloads to neighboring fog nodes, and managing communication with the cloud. Due to node mobility, workload variation, and energy depletion, the fog environment is inherently dynamic, making adaptive master head selection essential for maintaining quality of service. Finally, the cloud layer provides long-term storage of raw EEG data and fog responses for future monitoring, computationally intensive processing, and big data management.

A. Network Design

Let the set of EEG devices is denoted by $E = \{e_1, e_2, \dots, e_E\}$, where E represents the total number of EEG devices at base layer. Each EEG device $e_i \in E$ generates a collection of computational tasks

corresponding to EEG signal acquisition. The task set generated by e_1 is represented as $T_1 = \{t_1, t_2, \dots, t_{E1}\}$, and similarly the task set produced by the E -th EEG device is expressed as $T_E = \{t_1, t_2, \dots, t_{En}\}$. Where t_k denotes an individual EEG processing task and $E1$ and n indicate the total number of tasks generated by the corresponding EEG devices. In general, each device e_i produces a task set T_i and all tasks, $T = \bigcup_{i=1}^E T_i$, are transmitted to the edge gateway for aggregation and forwarding. Let the set of fog devices is given by $F = \{f_1, f_2, \dots, f_F\}$, where F denotes the total number of fog devices at service layer. The cloud layer is represented by C .

1) *Fog Delay Model*: In EEG-fog-IoT network, end-to-end latency critically affects the system responsiveness and decision-making process. Therefore, in this work, we consider four delay components - transmission delay, propagation delay, processing delay, and queuing delay.

Definition 1. *Transmission delay denotes the time required to transmit an EEG task from the edge gateway to the selected fog device. For task t_i of size s_i executed at fog node f_j , it is defined as*

$$\mathcal{D}_{f_j, t_i}^{\text{Trans}} = \frac{S(t_i)}{\mathcal{R}_{G \rightarrow f_j}^{\text{Wless}}} \quad (1)$$

Where $\mathcal{R}_{G \rightarrow f_j}^{\text{Wless}}$ represents the achievable wireless data rate between the gateway G and fog node f_j . Considering the wireless transmission medium in IoT network, the achievable data rate at time t is expressed as

$$\mathcal{R}_{G \rightarrow f_j}^{\text{Wless}}(t) = \mathcal{B} \log_2 \left(1 + \frac{\mathcal{P}_{G, f_j}^{\text{Chanl}}(t) \cdot \mathcal{P}^{\text{Trans}}}{\mathcal{P}^{\text{Noise}}} \right) \quad (2)$$

In Eq. (2), the channel power term $\mathcal{P}_{G, f_j}^{\text{Chanl}}(t)$ implicitly captures large-scale path loss and small-scale wireless variations between the edge gateway and fog node f_j as $\mathcal{P}_{G, f_j}^{\text{Chanl}}(t) = \kappa D_{G, f_j}^{-\alpha}$ [11]. Where, D_{G, f_j} denotes the Euclidean distance between gateway G and f_j , α represents the path-loss exponent determined by the propagation environment, and κ is a channel-dependent constant. In dynamic fog environment, both D_{G, f_j} and $\mathcal{P}_{G, f_j}^{\text{Chanl}}(t)$ vary over time due to fog count and node mobility.

Definition 2. *Propagation delay represents the signal travel time between the edge gateway and fog node over the wireless channel, expressed as*

$$\mathcal{D}_{G \rightarrow f_j}^{\text{Prop}} = \frac{D_{G, f_j}}{\mathcal{V}^{\text{Wless}}} \quad (3)$$

Where $\mathcal{V}^{\text{Wless}}$ is the propagation speed in wireless network medium. The master head selection includes propagation delay as one of the key components. It is because selecting a geographically closer fog node to an EEG device can substantially reduce latency.

Definition 3. *Processing delay is the computation time required to execute EEG task t_i on fog node f_j , given by*

$$\mathcal{D}_{f_j, t_i}^{\text{Proc}} = \frac{\mathcal{L}(t_i)}{\mathcal{F}_{f_j}^{\text{clk}}} \quad (4)$$

Where $\mathcal{L}(t_i)$ denotes the task length i.e. number of CPU cycles required by task t_i , and $\mathcal{F}_{f_j}^{\text{clk}}$ represents the processing capacity of fog node f_j . As the fog devices at the service layer are heterogeneous, hence the processing capabilities may vary in terms of clock frequency.

Definition 4. *Queuing delay refers to the waiting time of task t_i in the buffer of fog node f_j before execution. It is modeled as*

$$\mathcal{D}_{f_j, t_i}^{\text{Queu}} = \sum_{t_k \in Q_j} \mathcal{D}_{f_j, t_k}^{\text{Proc}} \quad (5)$$

Where Q_j denotes the set of tasks queued ahead of t_i at fog node f_j . Finally, the overall delay experienced by task t_i executed at fog node f_j is expressed as

$$\mathcal{D}_{f_j, t_i}^{\text{Total}} = \mathcal{D}_{f_j, t_i}^{\text{Trans}} + \mathcal{D}_{G \rightarrow f_j}^{\text{Prop}} + \mathcal{D}_{f_j, t_i}^{\text{Proc}} + \mathcal{D}_{f_j, t_i}^{\text{Queu}} \quad (6)$$

2) *Fog Energy Model*: In dynamic fog-based health-care IoT environments, energy efficiency is a critical factor due to the limited power resources of fog devices and the continuous arrival of EEG workloads. Each fog node consumes energy during wireless communication, task execution, and idle operation.

Definition 5. *The total energy consumed by fog node f_j while processing EEG tasks is modeled as the sum of communication energy, computation energy, and idle energy. i.e.*

$$\mathcal{E}_{f_j}^{\text{Total}} = \mathcal{E}_{f_j}^{\text{TR}} + \mathcal{E}_{f_j}^{\text{Proc}} + \mathcal{E}_{f_j}^{\text{Idle}} \quad (7)$$

Communication energy represents the energy consumption during wireless transmission and reception of EEG tasks by a fog device. It is directly proportional to the transmission delay and calculated as

$$\mathcal{E}_{f_j}^{\text{TR}} = 2 \cdot u_{f_j}^{\text{Trans}} \cdot \mathcal{D}_{f_j}^{\text{Trans}} \quad (8)$$

Where $u_{f_j}^{\text{Trans}}$ denotes the unit energy consumption per second for wireless communication and reception, and $\mathcal{D}_{f_j}^{\text{Trans}}$ is obtained from Eq. (1). The processing energy corresponds to the consumption energy expended while executing EEG tasks on fog device f_j , given by

$$\mathcal{E}_{f_j}^{\text{Proc}} = u_{f_j}^{\text{Proc}} \cdot \mathcal{D}_{f_j}^{\text{Proc}} \quad (9)$$

Where $u_{f_j}^{\text{Proc}}$ denotes the unit energy consumption for processing, and $\mathcal{D}_{f_j}^{\text{Proc}}$ is obtained from Eq. (4). Further, fog nodes consume baseline power even when no active tasks are being processed. Therefore, we consider idle energy $\mathcal{E}_{f_j}^{\text{Idle}}$ which represents static power draw due to memory, peripherals, and system clocks [4].

III. PROPOSED METHODOLOGY

A. Problem Formulation

In our healthcare IoT network, multiple heterogeneous fog devices collaboratively process latency-sensitive biomedical tasks. Due to dynamic fog participation, and varying computational workloads, selecting an optimal MH becomes a adaptive decision problem. The objective of this work is to dynamically select a master head fog device that jointly minimizes communication-computation latency and energy expenditure. This is formulated as,

$$\mathbb{F}^* \leftarrow \arg \min_{f_j \in F} \left\{ \sum_{t_i \in T} \mathcal{D}_{f_j, t_i}^{\text{Total}}, \mathcal{E}_{f_j}^{\text{Total}} \right\} \quad (10)$$

Due to heterogeneous fog resources and continuously changing network states, no single fog device can simultaneously achieve global optimality across both objectives [12]. Therefore, in this work, we adopt Pareto optimality to identify a set of non-dominated fog candidates with the following constraints - $\mathcal{F}_{f_j}^{\text{clk}} > 0$, and $\mathcal{E}_{f_j}^{\text{Idle}} < \mathcal{E}_{f_j}^{\text{Min}}$.

B. Proposed Master Head Selection

The selection procedure consists of the following steps:

- **Step 1:** For each fog device $f_i \in F$, the total delay and total energy are computed using Eq. (4) and (7) respectively. These values quantify the latency and energy impact of selecting MH.
- **Step 2:** Since delay and energy possess different scales, max-min normalization is applied to map each objective into the range $[0, 1]$ as,

$$\hat{x}_{f_j} = \frac{x_{f_j} - x_{\min}}{x_{\max} - x_{\min}}, \quad (11)$$

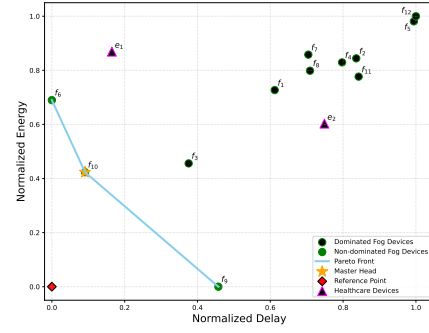


Fig. 2. Shows a scenario of proposed MH selection for $F = 12$ and $E = 2$.

where $x_{\min}$ and $x_{\max}$ denote the minimum and maximum values of the objective across all fog devices, respectively. Here, x_{f_j} represents either the total delay $\mathcal{D}_{f_j}^{\text{Total}}$ or the total energy $\mathcal{E}_{f_j}^{\text{Total}}$. The objective normalization ensures fair comparison among heterogeneous fog devices and prevents any single metric from dominating the Pareto optimization process.

- **Step 3:** Each fog device is pairwise compared with others to evaluate Pareto dominance. A fog device f_a dominates f_b if it exhibits no worse performance in both objectives and strictly better performance in at least one.
- **Step 4:** Based on dominance relationships with at least one strict inequality, fog devices are classified into: non-dominated set F^+ and dominated set F^- . The set F^+ represents feasible master head candidates offering optimal delay-energy tradeoffs.
- **Step 5:** A hypothetical Reference Point (RP) is constructed using the minimum normalized delay and minimum normalized energy among all fog devices. For each $f_i \in F^+$, its Euclidean distance to the RP is computed. This measures proximity to the ideal operating condition.
- **Step 6:** Finally, the fog device within F^+ having the minimum distance to the RP is selected as the MH. This ensures balanced optimization of both latency and energy objectives.

Additionally we include a fallback mechanism for scenarios, where $F^+ = \emptyset$, the MH is selected based on Fog Computational Efficiency (FCE).

Definition 6. Fog Computational Efficiency of j^{th} fog device is defined as the average number of clock cycles per instruction of processor's performance, and is

Algorithm 1 Proposed Dynamic MH Selection

Require: Fog device set F , EEG task set T **Ensure:** Selected optimal Master Head (MH)

```
1: Initialize  $F^+ \leftarrow \emptyset$ ,  $F^- \leftarrow \emptyset$ 
2: for each fog device  $f_j \in F$  do
3:   Compute total delay  $\mathcal{D}_{f_j}^{\text{Total}}$  using Eq. (4)
4:   Compute total energy  $\mathcal{E}_{f_j}^{\text{Total}}$  using Eq. (7)
5: end for
6: Apply max-min normalization to  $\mathcal{D}_{f_j}^{\text{Total}}$  and  $\mathcal{E}_{f_j}^{\text{Total}}$ 
7: for each pair  $(f_a, f_b)$  where  $f_a, f_b \in F$  and  $a \neq b$ 
   do
8:   if  $f_a$  dominates  $f_b$  then
9:     Assign  $f_b$  to  $F^-$ 
10:    Assign  $f_a$  to  $F^+$ 
11:   end if
12: end for
13: Update  $F^+$ 
14: if  $|F^+| = 1$  then
15:   MH  $\leftarrow$  the only element in  $F^+$ 
16:   return MH
17: end if
18: if  $F^+ \neq \emptyset$  then
19:   Find Reference Point (RP)
20:   for each  $f_j \in F^+$  do
21:     Compute Euclidean distance to RP
22:   end for
23:   MH  $\leftarrow$  fog device with minimum distance to RP
24: else
25:   for each  $f_j \in F$  do
26:     Compute  $FCE_{f_j} = \frac{\zeta_{f_j} \times \mathcal{F}_{f_j}^{\text{clk}}}{\mathcal{M}_{f_j} \times 10^6}$ 
27:   end for
28:   MH  $\leftarrow$  fog device with minimum FCE
29: end if
30: return MH
```

calculated as

$$FCE = \frac{\zeta_{f_j} \times \mathcal{F}_{f_j}^{\text{clk}}}{\mathcal{M}_{f_j} \times 10^6} \quad (12)$$

where ζ_{f_j} denotes number of cores associated with fog processor and $\mathcal{M}_{f_j}$ denotes MIPS of the processor.

The lower FCE indicates higher processing efficiency and faster task execution. The workflow of the proposed mechanism is outlined in Algorithm 1.

C. Illustrative Example

In this section, we demonstrate the proposed master head selection strategy, considering a dynamic fog en-

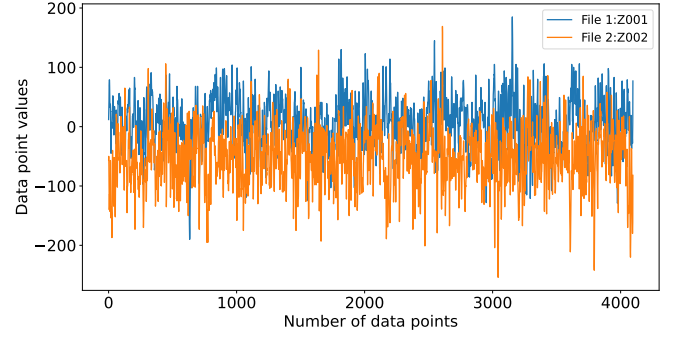


Fig. 3. Shows dataset representation of two EEG files.

vironment consisting of 12 heterogeneous fog devices with varying computational and energy characteristics from [5] [13]. This heterogeneity ensures that no single fog device is inherently biased toward selection based on its specification. Moreover, Fig. 2 illustrates the distribution of fog devices in the normalized delay-energy objective space. After applying the proposed dominance-based classification, we get $F^+ = \{f_6, f_9, f_{10}\}$. Among them, f_{10} exhibits the minimum distance to the RP and is therefore selected as the MH. This selection is further justified as a set of fog devices possesses higher values in a few criteria than f_{10} but f_{10} achieves the best tradeoff between latency and energy among all candidates. This validates that the selection is not driven by a single metric but emerges from balanced multi-objective optimization.

IV. RESULT ANALYSIS

The simulation is carried out using 64 bits WIN OS i5 processor having 8 GB RAM and over 100 iterations. We consider $F = 12$, $E = 2$, $T = 100$, and $\mathcal{B} = 10$ Mbps for experimental set up. Our work is demonstrated using three metrics - average queuing delay, average total delay, and total energy consumption. Later, we compare the results against conventional cloud processing and static fog environment [14]. Fig. 3 shows EEG dataset of two files - Z001 and Z002, consisting of 4096 data points in each file [15].

Fig. 4 illustrates the variation of average queuing delay and average total delay with respect to increasing task arrivals under different fog configurations (2 – 10 fog devices). As expected, both metrics exhibit a monotonic increase with workload intensity due to processing demand. However, a clear performance gain is observed as the number of participating fog devices increases. For two fog devices, the average queuing delay rises sharply

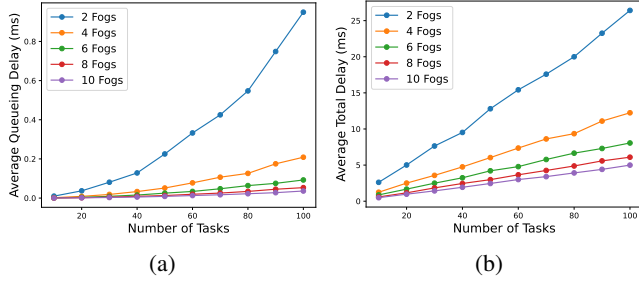


Fig. 4. Shows performance analysis of proposed work for average queuing delay and average total delay as metrics.

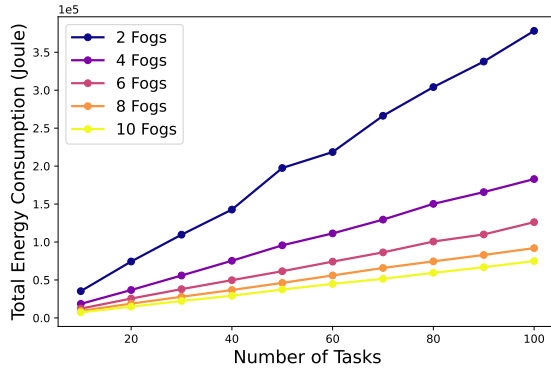


Fig. 5. Shows analysis of energy consumption for increasing fog devices and workload.

from 0.37 ms at 20 tasks to nearly 0.95 ms at 100 tasks. In contrast, with ten fog devices, the queuing delay remains below 0.036 ms even at maximum workload. This significant improvement confirms that distributing EEG tasks across multiple fog nodes effectively mitigates buffer congestion and reduces waiting time. Similarly, average total delay increases from about 5 ms to nearly 26.41 ms for two fog devices as the task count grows from 20 to 100. For ten fog devices, it is constrained within 0.97 – 4.98 ms over the same workload range. This behavior validates the effectiveness of the proposed Pareto-based master head selection strategy in dynamically distributing workloads among heterogeneous fog nodes.

Fig. 5 presents the variation of total energy consumption with increasing number of EEG tasks under different fog configurations. With only two fog nodes, the energy consumption rises steeply from approximately 0.74×10^5 J at 20 tasks to nearly 3.78×10^5 at 100 tasks. On the other side, this remains within $[0.14 - 0.75] \times 10^5$ J over the same workload range for ten fog devices. The average energy consumption increases with an increase

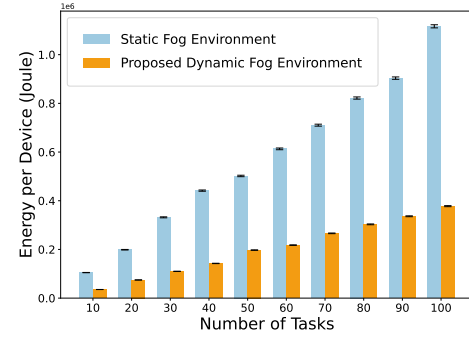


Fig. 6. Shows comparative analysis of energy consumption per device with different task sets between static fog environment and the proposed dynamic environment.

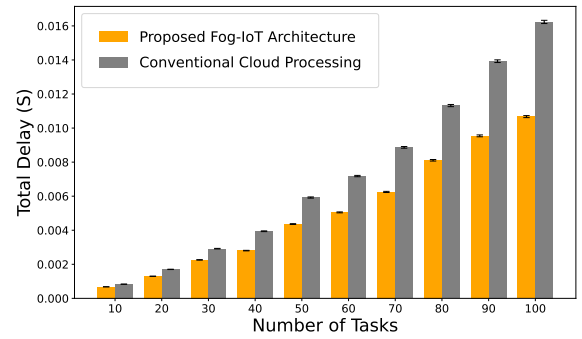


Fig. 7. Shows comparative analysis of total delay varying number of tasks between conventional cloud-centric architecture and the proposed fog-IoT architecture.

in number of tasks but the trend is reverse with an increase in number of fog devices, as expected. This confirms that our method significantly improves energy sustainability along with minimum delay by dynamically coordinating fog resources.

Furthermore, we compare our work with two baseline methods such as centralized cloud-based processing and static fog-IoT network. Fig. 6 compares the per-device energy consumption between a conventional static fog environment and the proposed dynamic fog framework for minimum fog configuration. As the number of tasks increases, energy consumption grows rapidly from approximately 1.05×10^5 at 10 tasks to nearly 11.13×10^5 at 100 tasks. In contrast, the proposed dynamic fog environment maintains significantly lower energy levels, ranging from about 0.35×10^5 to 3.78×10^5 over the same workload. This corresponds to an average energy reduction of nearly 60 – 70% per device at higher task volumes. Additionally, Fig. 7 presents a comparison of total end-to-end delay between the proposed fog-assisted

IoT architecture and a conventional cloud-centric approach. The cloud-based model exhibits steadily increasing latency, reaching approximately 0.016 s at 100 tasks compared to 0.01 s in proposed method. This indicates to an improvement to a delay reduction of about maximum 35%. These findings confirm that integrating fog computing with dynamic master head selection substantially outperforms cloud-centric architectures in latency-sensitive healthcare scenarios.

V. CONCLUSION

This paper presented a Pareto-based dynamic master head selection framework for EEG-driven healthcare IoT networks operating in heterogeneous fog environments. A three-layer architecture comprising EEG devices, fog nodes, and cloud infrastructure was designed to enable low-latency signal processing while ensuring energy-efficient resource utilization. The master head selection problem was formulated that jointly minimizes end-to-end delay and fog energy consumption. Extensive simulation results demonstrated that the proposed approach significantly reduces queuing delay, total latency, and per-device energy consumption compared to static fog and cloud-centric architectures. In particular, delay improvements of up to 35% and energy savings approaching 60 – 70% were achieved under increasing task loads, validating the effectiveness of dynamic fog coordination. As future work, this study will be extended to incorporate GenAI-assisted workload forecasting, and adaptive task scheduling mechanisms to further improve responsiveness and resilience in large-scale fog-based healthcare deployments.

REFERENCES

- [1] S. Vaid, P. Singh, and C. Kaur, "Eeg signal analysis for bci interface: A review," in *2015 fifth international conference on advanced computing & communication technologies*. IEEE, 2015, pp. 143–147.
- [2] V. A. S, N. E, and V. G. R, "Performance evaluation of mobile edge computing using 5g networks," in *2021 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, 2021, pp. 1–6.
- [3] X. Wang and Y. Wu, "Fog-assisted internet of medical things for smart healthcare," *IEEE Transactions on Consumer Electronics*, vol. 69, no. 3, pp. 391–399, 2023.
- [4] E. S. S. Kaushal, R. K. Rath, S. Sreeja, and A. Hazra, "3d-pos: 3d pareto optimized head selection for fog-enabled smart eeg healthcare iot," in *2025 IEEE 6th India Council International Subsections Conference (INDISCON)*. IEEE, 2025, pp. 1–6.
- [5] S. H. Kurra, R. K. Rath, and S. Sreeja, "2dp-fhs: 2d pareto optimized fog head selection for multiple eeg healthcare data analysis and computations," in *International Conference on Advances in Computing and Data Sciences*. Springer, 2024, pp. 58–68.
- [6] L. S. Vidyaratne and K. M. Iftekhharuddin, "Real-time epileptic seizure detection using eeg," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 11, pp. 2146–2156, 2017.
- [7] S. K. Satapathy, S. Pattnaik, and R. Rath, "Automated sleep staging classification system based on convolutional neural network using polysomnography signals," in *2022 IEEE Delhi section conference (DELCON)*. IEEE, 2022, pp. 1–10.
- [8] T. R. Mullen, C. A. Kothe, Y. M. Chi, A. Ojeda, T. Kerth, S. Makeig, T.-P. Jung, and G. Cauwenberghs, "Real-time neuroimaging and cognitive monitoring using wearable dry eeg," *IEEE transactions on biomedical engineering*, vol. 62, no. 11, pp. 2553–2567, 2015.
- [9] S. Chakraborty, S. Bhowmick, P. Talaga, and D. P. Agrawal, "Fog networks in healthcare application," in *2016 IEEE 13th international conference on mobile ad hoc and sensor systems (MASS)*. IEEE, 2016, pp. 386–387.
- [10] R. K. Rath, S. Sreeja, A. Hazra, and R. Nanda, "A lossless healthcare data compression approach using near-edge computing," in *Industry 5.0: Key Technologies and Drivers*. Springer, 2025, pp. 69–92.
- [11] R. Siyadatzadeh, F. Mehrafrooz, M. Ansari, B. Safaei, M. Shafique, J. Henkel, and A. Ejlali, "Relief: A reinforcement-learning-based real-time task assignment strategy in emerging fault-tolerant fog computing," *IEEE Internet of Things Journal*, vol. 10, no. 12, pp. 10 752–10 763, 2023.
- [12] A. Bandopadhyay, S. Swain, R. Singh, P. Sarkar, S. Bhattacharyya, and L. Mrsic, "Game-theoretic resource allocation and dynamic pricing mechanism in fog computing," *IEEE Access*, vol. 12, pp. 51 704–51 718, 2024.
- [13] R. Nanda, P. Sakthivel, R. K. Rath, and A. Hazra, "An efficient master head selection for multi-eeg to multi-fog iot network using 6g-driven faas," *Computer Communications*, p. 108429, 2026.
- [14] A. Patel, R. Nanda, and R. K. Rath, "Profs: Pareto-based resource-aware optimized fog head selection in healthcare iot systems," in *2025 IEEE 3rd International Symposium on Sustainable Energy, Signal Processing and Cybersecurity*, 2025, pp. 1–5.
- [15] R. G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, and C. E. Elger, "Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state," *Physical Review E*, vol. 64, no. 6, p. 061907, 2001.